the clues are likely to lie in deeper and more detailed studies of the early universe, when that asymmetry first occurred.

Wherever we end up looking for data, in the quest for a theory of everything, we have two complementary approaches. One is to examine the phenomena we already see in nature that don't fit into established physical theories, so we can build new and better theories to explain them. The other is to just try to break the theories we have—write down hypothetical extreme cases that might not have been tested yet, and see if we can find a new way to look at the data that will show us if the theory still works there. A combination of these two approaches is pretty much always how we move forward in physics. It's how we went from Newtonian gravity, which works extremely well in everyday situations, to Einstein's general relativity. GR would be massive overkill for a block sliding down an inclined plane, but is absolutely essential for explaining the bending of light around extremely massive objects in space or the tiny shifts in the orbit of Mercury deep in the gravitational well of the Sun.

Newtonian gravity had to be replaced so we could move on to the superior general relativity; now it's general relativity's turn to be supplanted by the next big thing.

But GR has been so resistant to these efforts, we might end up having to rearrange the entire universe instead.

MAKING SPACE

There's a classic episode of *Star Trek: The Next Generation* in which, through a complicated series of events, Dr. Crusher ends up being the only person on the ship while it's trapped in some kind of weird hazy bubble. So many strange things are happening, including the sudden disappearance of the rest of the crew, and it's all at such odds with the readings of the sensors, her medical expertise tells her there's a very good chance she's hallucinating. But when her medical diagnostic fails to come up with any problem, she reaches the next logical conclusion: "If there's nothing wrong with me," she says, "maybe there's something wrong with the universe!" As it happens (and, apologies for the spoiler, but the episode aired in 1990; you've had three decades to watch it), she's absolutely right.

For a while now, some physicists have suspected that the incongruous weakness of gravity might be forcing them to a similar conclusion. Maybe